

DBC2 150W THERMAL TESTS SAMPLE UNIT 4

**18.05.2023
ENKO**

All thermal tests held down with the sample product according with this specs:

- 150W Sample
- 15VDC 10A Configuration.
- 230VAC line voltage.
- With the closed environment.
- All tests were started in a room temperature.

Test specifics:

Test 1:

- Unit mounted on a rack upside down. (AC line was on downside and DC side was on upside.)
- Black box of the unit itself was closed.
- Measurements were made from outside of the box with a thermal camera.



Image 1 – Wrong mounted test setup. (From right: AC Power meter, IDC, VDC)

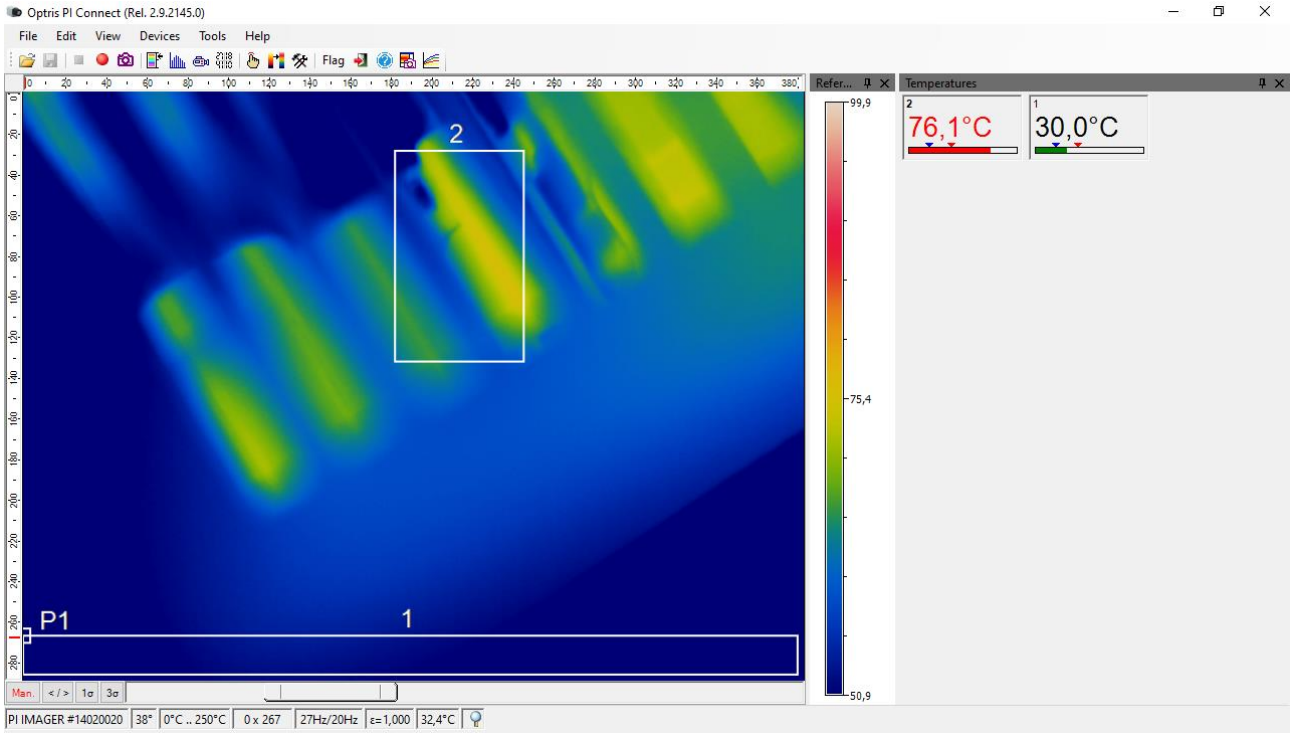


Image 2 – Wrong mounted, Secunder MOSFET.

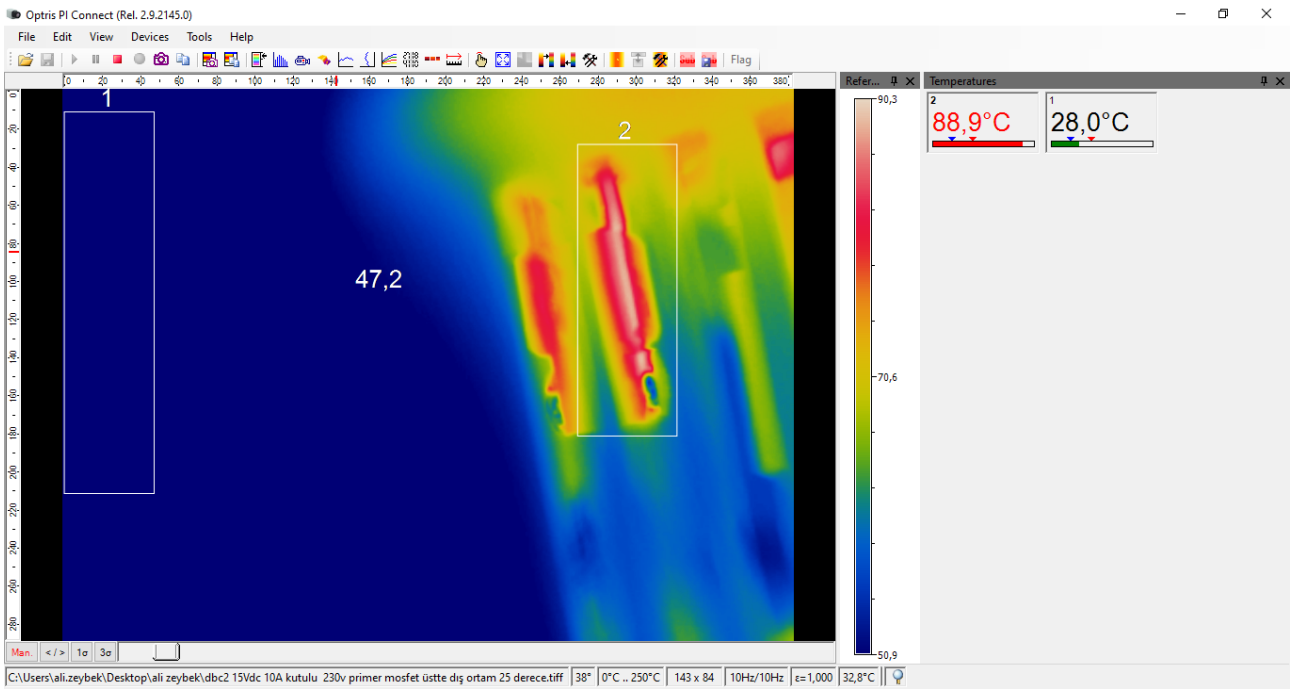


Image 3 – Wrong mounted, Primer MOSFET.

Test 2:

- Unit mounted on a rack right side up. (AC line was on upside and DC side was on downside.)
- Black box of the unit itself was closed.
- Measurements were made from outside of the box with a thermal camera.

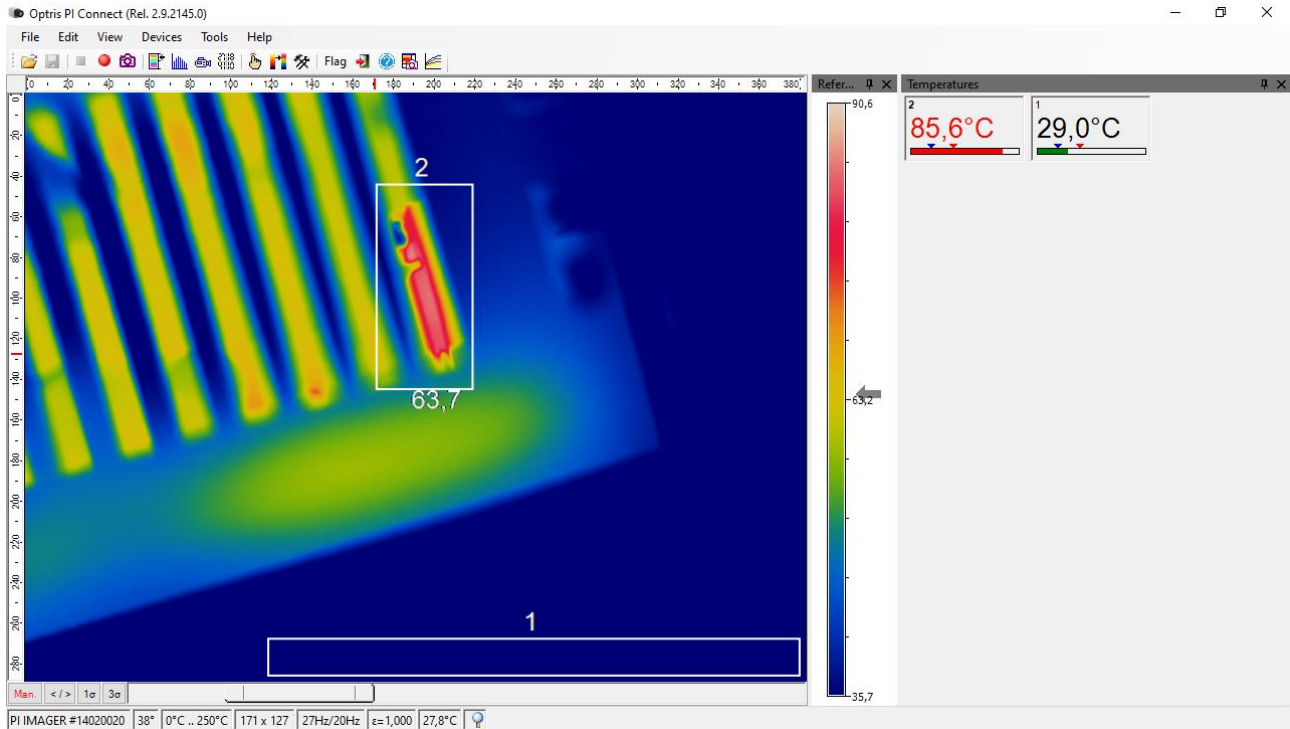


Image 4 – Normal mounted Primer MOSFET.

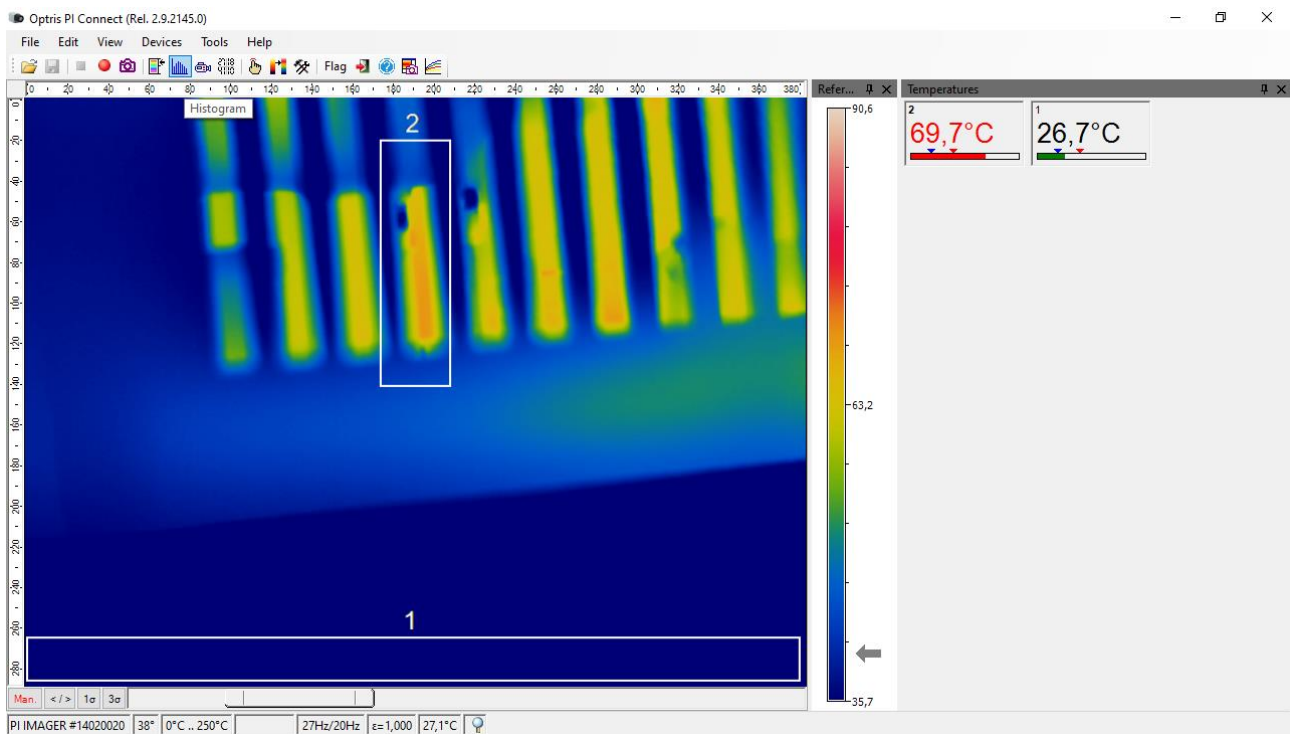


Image 5 – Normal mounted Secunder MOSFET.

Test 3:

- Unit mounted on a rack upside down. (AC line was on downside and DC side was on upside.)
- Box of the unit was open.
- Measurements were made from outside of the box with a thermal camera.

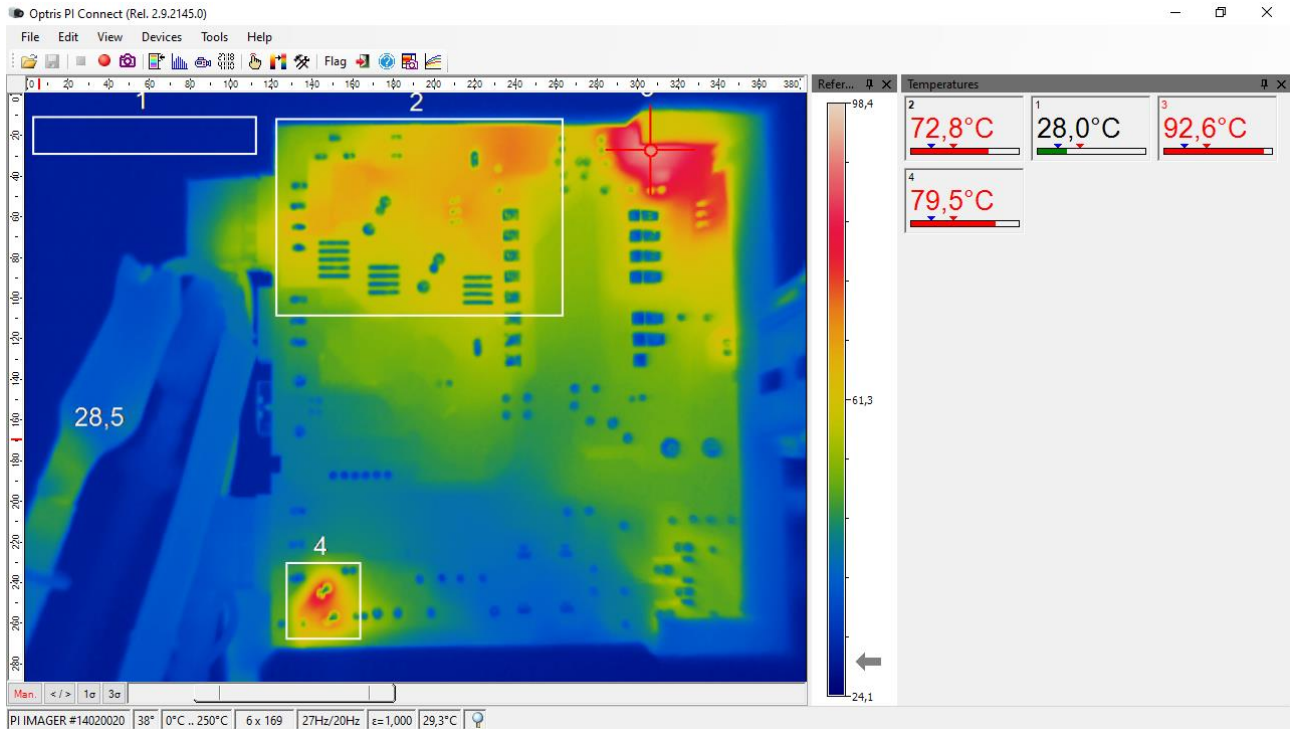


Image 6 – Wrong mounted Backside of PCB. (4: NTC PCB, 2: Secnder PCB, 3 Pri Snubber PCB)

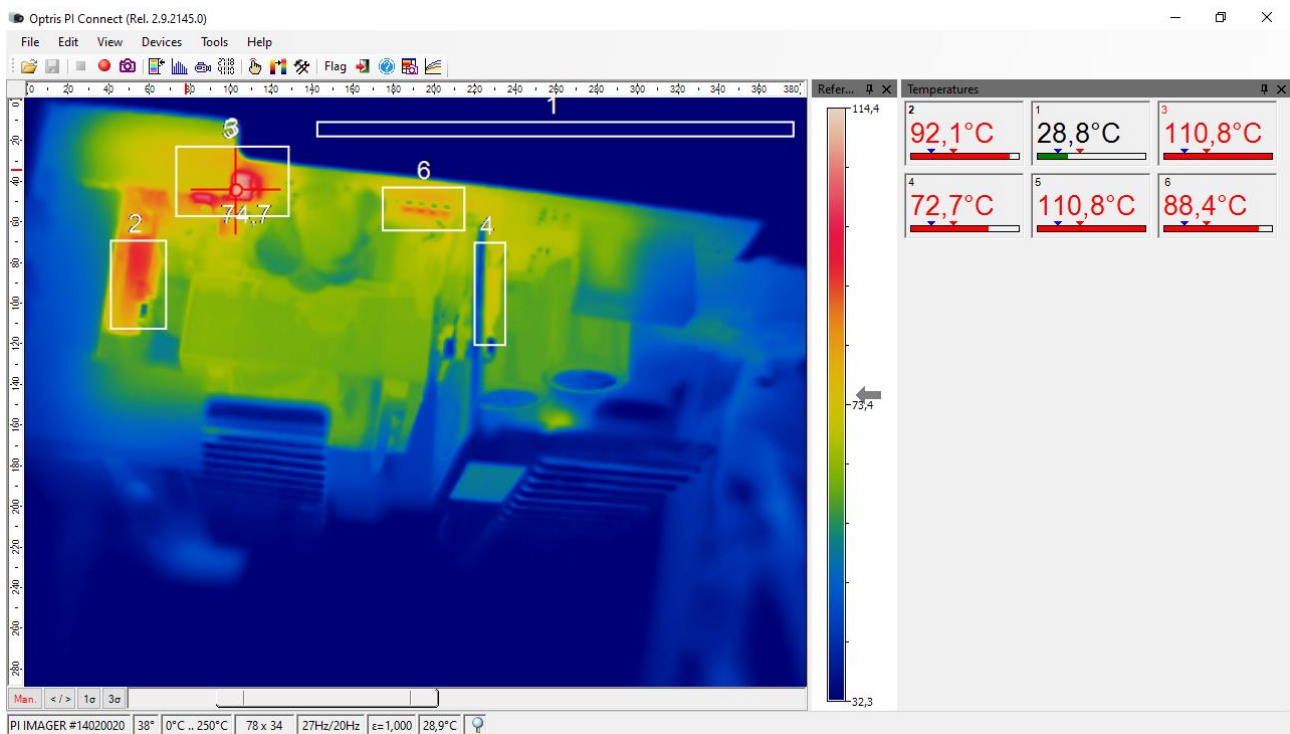


Image 7 – Wrong mounted, MOSFETs. (From right: Pri MOS, Pri Snubbers, Sec Snubbers, Sec MOS)

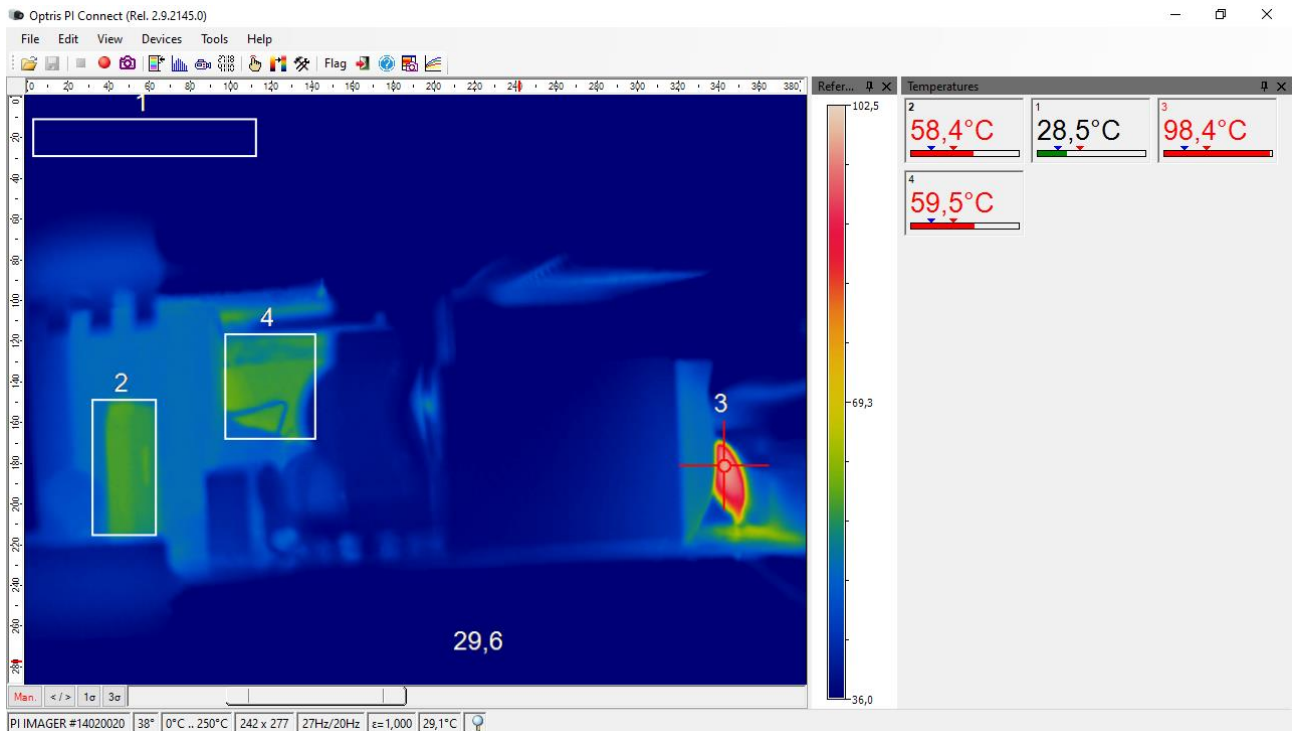


Image 8 – Wrong mounted, NTC. (From right Bridge, Pri MOS, NTC)

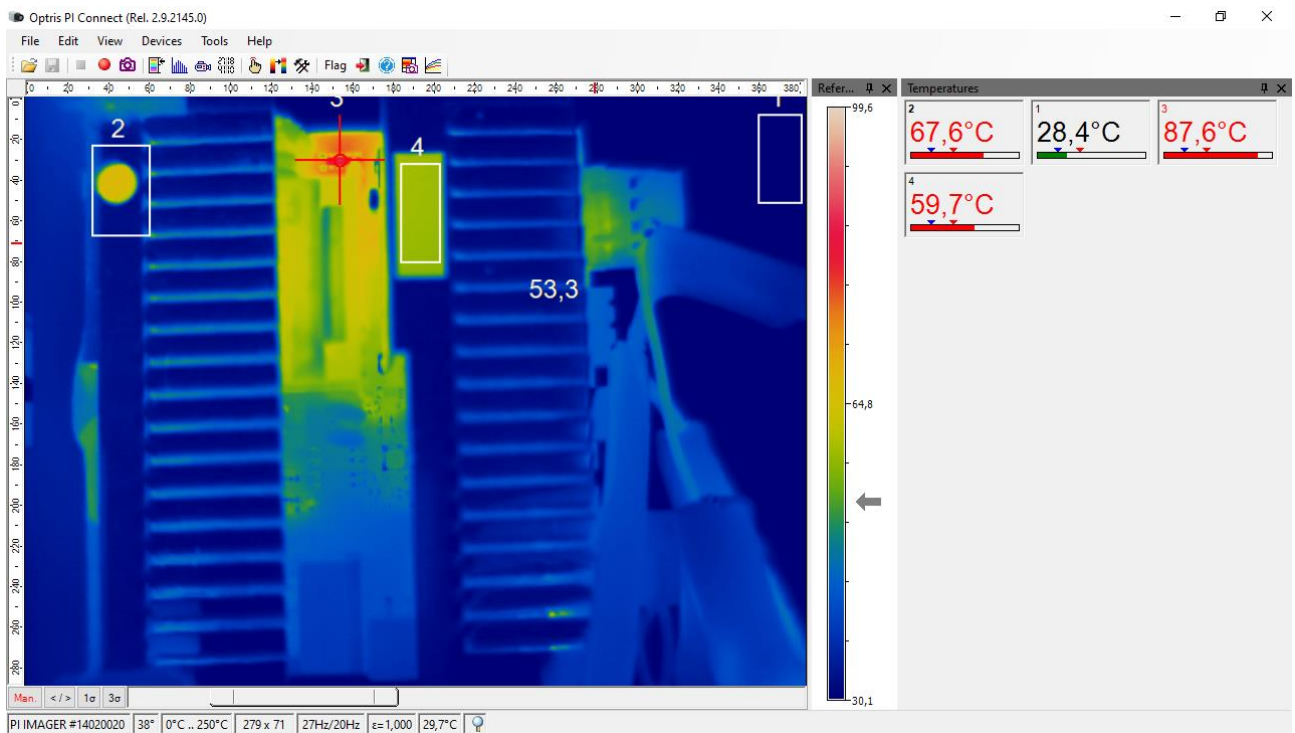


Image 9 – Wrong mounted, Snubbers. (From right: 2: Pri heatsink, 3: Pri snubbers, 4: Secnder heatsink)

Test 4:

- Unit mounted on a rack right. (AC line was on upside and DC side was on downside.)
- Black box of the unit itself was closed.
- Measurements were made from outside of the box with a thermal camera.

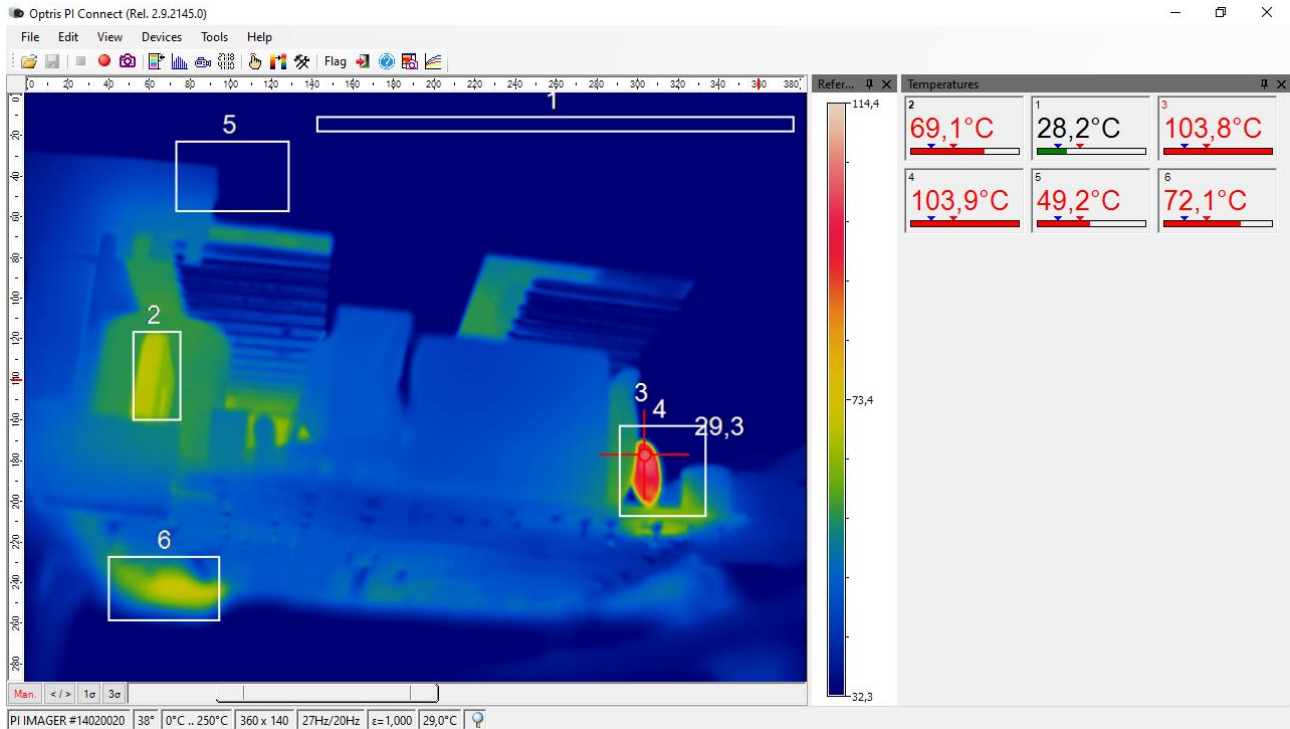


Image 10 – Normal mounted, Bridge and NTC. (From right: 6: Pri snubber PCB, 2: Bridge, 3 and 4: NTC)

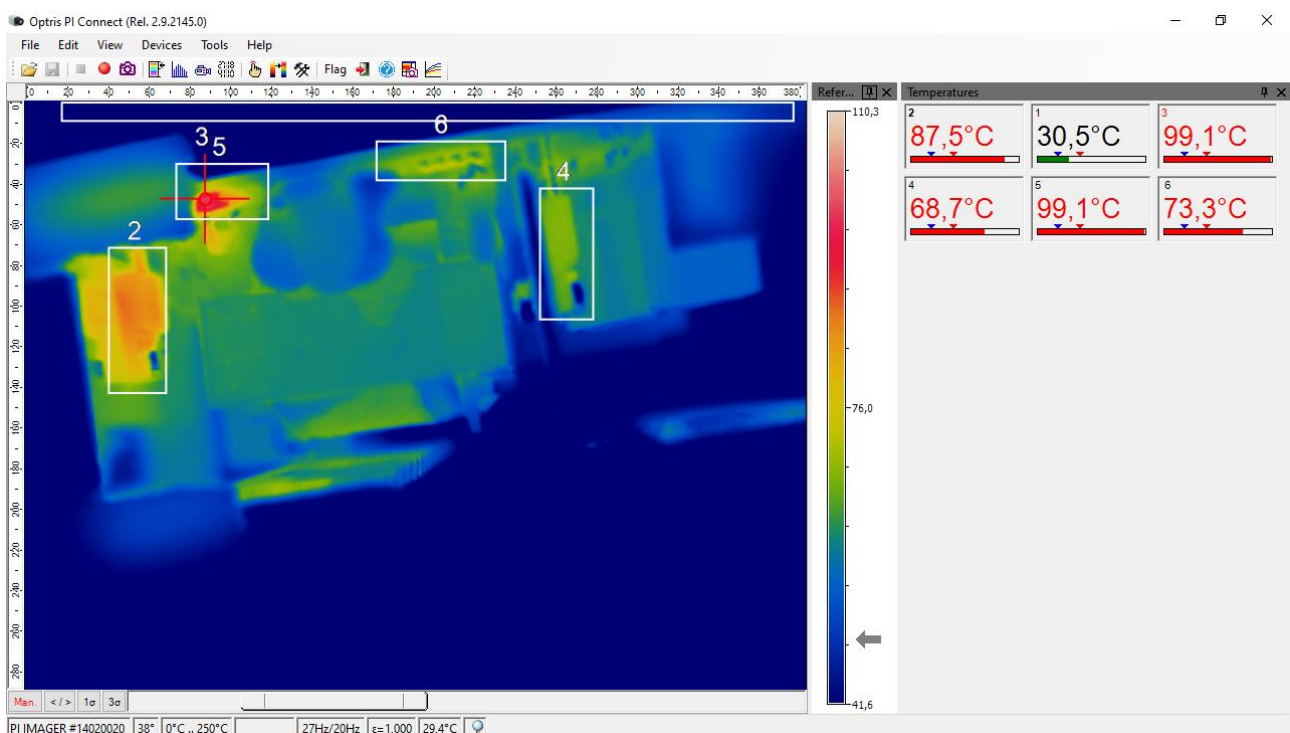


Image 11 – Normal mounted, MOSFETs and Snubbers. (From right: Pri MOS, Pri Snubbers, Sec snubbers, Sec MOS.)



Image 12 – Wrong mounted open test setup. (From right: AC Power meter, IDC, VDC)

Test Number	Pri MOSFET (°C)	Sec MOSFET (°C)	Bridge Rect. (°C)	Pri Snubber(°C)	NTC PCB(°C)	Pri Snubber PCB(°C)	Environment (°C)
Test 1	88,9	76,1	-	-	-	-	27
Test 2	85,6	69,7	-	-	-	-	28
Test 3	92,1	72,7	58,4	110,8	98,4	-	28
Test 4	87,5	68,7	69,1	99,1	103,8	72,1	29